

Chapter 2: Solid-State Electronics - Detailed Problem Solutions

Compact LaTeX version. Constants used unless stated otherwise: $q = 1.602 \times 10^{-19} \text{ C}$, $V_T \approx 0.025 \text{ V}$ at 300 K, $n_i \approx 10^{10} \text{ cm}^{-3}$ for Si at room temperature.

Problem 2.1 Given $\rho_{Al} = 2.83 \mu\Omega \cdot \text{cm} = 2.83 \times 10^{-6} \Omega \cdot \text{cm}$. Conductors have very small resistivity, typically below about $10^{-3} \Omega \cdot \text{cm}$. Since

$$2.83 \times 10^{-6} < 10^{-3},$$

pure aluminum is a **conductor**.

Problem 2.2 Given $\rho_{SiO_2} = 10^{15} \Omega \cdot \text{cm}$. Insulators have very large resistivity, typically above about $10^5 \Omega \cdot \text{cm}$. Since

$$10^{15} \gg 10^5,$$

silicon dioxide is an **insulator**.

Problem 2.3 The current density limit is

$$J_{max} = 10 \text{ MA/cm}^2 = 10^7 \text{ A/cm}^2.$$

The cross-sectional area is

$$A = (5 \mu\text{m})(1 \mu\text{m}) = 5 \mu\text{m}^2 = 5 \times 10^{-8} \text{ cm}^2.$$

Thus

$$I_{max} = J_{max} A = (10^7)(5 \times 10^{-8}) = 0.50 \text{ A}.$$

$$I_{max} = 500 \text{ mA}.$$

Problem 2.4 The interconnect runs diagonally across a $20 \text{ mm} \times 20 \text{ mm}$ die:

$$L = \sqrt{20^2 + 20^2} \text{ mm} = 28.28 \text{ mm} = 2.828 \text{ cm}.$$

For aluminum, $\rho = 2.82 \times 10^{-6} \Omega \cdot \text{cm}$ and $R = \rho L/A$.

(a) If $t = 1 \mu\text{m}$ and $w = 5 \mu\text{m}$,

$$A = 5 \mu\text{m}^2 = 5 \times 10^{-8} \text{ cm}^2, \quad R = \frac{(2.82 \times 10^{-6})(2.828)}{5 \times 10^{-8}} = 1.60 \times 10^2 \Omega.$$

$$R \approx 160 \Omega.$$

(b) If $t = 0.5 \mu\text{m}$,

$$A = 2.5 \times 10^{-8} \text{ cm}^2, \quad R = \frac{(2.82 \times 10^{-6})(2.828)}{2.5 \times 10^{-8}} = 3.19 \times 10^2 \Omega.$$

$$R \approx 319 \Omega.$$

Problem 2.5 Repeat Prob. 2.4 with copper, $\rho_{Cu} = 1.66 \times 10^{-6} \Omega \cdot \text{cm}$.

(a) For $A = 5 \times 10^{-8} \text{ cm}^2$,

$$R = \frac{(1.66 \times 10^{-6})(2.828)}{5 \times 10^{-8}} = 93.9 \Omega.$$

(b) For $A = 2.5 \times 10^{-8} \text{ cm}^2$,

$$R = \frac{(1.66 \times 10^{-6})(2.828)}{2.5 \times 10^{-8}} = 188 \Omega.$$

Copper has lower resistance because its resistivity is lower than aluminum's.

Problem 2.6 Use

$$n_i^2 = BT^3 \exp\left(-\frac{E_G}{kT}\right), \quad k = 8.62 \times 10^{-5} \text{ eV/K}.$$

For Si, $B = 1.08 \times 10^{31} \text{ K}^{-3} \text{ cm}^{-6}$, $E_G = 1.12 \text{ eV}$. For Ge, $B = 2.31 \times 10^{30} \text{ K}^{-3} \text{ cm}^{-6}$, $E_G = 0.66 \text{ eV}$.

Material	77 K	300 K	500 K
Si	$2.01 \times 10^{-10} \text{ cm}^{-3}$	$6.73 \times 10^9 \text{ cm}^{-3}$	$8.36 \times 10^{13} \text{ cm}^{-3}$
Ge	35.9 cm^{-3}	$2.27 \times 10^{13} \text{ cm}^{-3}$	$8.04 \times 10^{15} \text{ cm}^{-3}$

Problem 2.7 Solve the silicon expression

$$n_i^2 = 1.08 \times 10^{31} T^3 \exp\left(-\frac{1.12}{8.62 \times 10^{-5} T}\right)$$

numerically.

(a) For $n_i = 10^{13} \text{ cm}^{-3}$, $n_i^2 = 10^{26}$, giving $T \approx 506 \text{ K}$.

(b) For $n_i = 10^{15} \text{ cm}^{-3}$, $n_i^2 = 10^{30}$, giving $T \approx 739 \text{ K}$.

Problem 2.8 For GaAs, use $B = 1.27 \times 10^{29} \text{ K}^{-3} \text{ cm}^{-6}$ and $E_G = 1.42 \text{ eV}$:

$$n_i^2 = 1.27 \times 10^{29} T^3 \exp\left(-\frac{1.42}{8.62 \times 10^{-5} T}\right).$$

T	n_i
300 K	$2.21 \times 10^6 \text{ cm}^{-3}$
100 K	$6.03 \times 10^{-19} \text{ cm}^{-3}$
450 K	$3.82 \times 10^{10} \text{ cm}^{-3}$

Problem 2.9 Set $n_i = 10^{10} \text{ cm}^{-3}$. Then

$$10^{20} = 1.08 \times 10^{31} T^3 \exp\left(-\frac{1.12}{8.62 \times 10^{-5} T}\right).$$

Solving numerically gives $T = 305.2 \text{ K}$.

Problem 2.10 Use

$$v_n = -\mu_n E, \quad v_p = \mu_p E.$$

With $E = +2500 \text{ V/cm}$, $\mu_n = 700 \text{ cm}^2/(\text{V} \cdot \text{s})$, $\mu_p = 250 \text{ cm}^2/(\text{V} \cdot \text{s})$,

$$v_n = -700(2500) = -1.75 \times 10^6 \text{ cm/s}, \quad v_p = 250(2500) = 6.25 \times 10^5 \text{ cm/s}.$$

Current densities:

$$j_n = -qn v_n = -(1.602 \times 10^{-19})(10^{17})(-1.75 \times 10^6) = 2.80 \times 10^4 \text{ A/cm}^2,$$

$$j_p = qp v_p = (1.602 \times 10^{-19})(10^3)(6.25 \times 10^5) = 1.00 \times 10^{-10} \text{ A/cm}^2.$$

Problem 2.11 Use $|j| = qNv$. With $v_{max} = 10^7 \text{ cm/s}$,

$$|j_n| = (1.602 \times 10^{-19})(10^{18})(10^7) = 1.60 \times 10^6 \text{ A/cm}^2,$$

$$|j_p| = (1.602 \times 10^{-19})(10^2)(10^7) = 1.60 \times 10^{-10} \text{ A/cm}^2.$$

The total current density is dominated by electrons: $j_T \approx 1.60 \times 10^6 \text{ A/cm}^2$.

Problem 2.12 Use $j = Qv$, so

$$v = \frac{j}{Q} = \frac{-2000 \text{ A/cm}^2}{0.01 \text{ C/cm}^3} = -2.00 \times 10^5 \text{ cm/s}.$$

The negative sign means the carrier velocity is in the negative reference direction.

Problem 2.13 The charge density is $Q = 0.4 \text{ C/cm}^3$ and $v_{max} = 10^7 \text{ cm/s}$. Thus

$$j_{max} = Qv_{max} = 0.4(10^7) = 4.0 \times 10^6 \text{ A/cm}^2 = 4.0 \text{ MA/cm}^2.$$

Problem 2.14 With $E = -2000 \text{ V/cm}$,

$$v_n = -\mu_n E = -1000(-2000) = 2.00 \times 10^6 \text{ cm/s},$$

$$v_p = \mu_p E = 400(-2000) = -8.00 \times 10^5 \text{ cm/s}.$$

Then

$$j_n = -qn v_n = -(1.602 \times 10^{-19})(10^3)(2.00 \times 10^6) = -3.20 \times 10^{-10} \text{ A/cm}^2,$$

$$j_p = qp v_p = (1.602 \times 10^{-19})(10^{17})(-8.00 \times 10^5) = -1.28 \times 10^4 \text{ A/cm}^2.$$

Problem 2.15

(a) $10\text{ }\mu\text{m} = 10^{-3}\text{ cm}$, so

$$E = \frac{V}{L} = \frac{5}{10^{-3}} = 5.0 \times 10^3 \text{ V/cm}.$$

(b) If $E_{max} = 10^5 \text{ V/cm}$,

$$V_{max} = E_{max}L = (10^5)(10^{-3}) = 100 \text{ V}.$$

Problem 2.16 For holes,

$$j_{p,max} = qpv_{max} = (1.602 \times 10^{-19})(10^{19})(10^7) = 1.60 \times 10^7 \text{ A/cm}^2.$$

The area is

$$A = (1\text{ }\mu\text{m})(25\text{ }\mu\text{m}) = 25 \times 10^{-8} \text{ cm}^2 = 2.5 \times 10^{-7} \text{ cm}^2.$$

Therefore

$$I_{max} = jA = (1.60 \times 10^7)(2.5 \times 10^{-7}) = 4.00 \text{ A}.$$

Problem 2.17 For intrinsic silicon,

$$\sigma = qn_i(\mu_n + \mu_p).$$

The insulator boundary is approximately $\sigma = 10^{-5}(\Omega \cdot \text{cm})^{-1}$. With $\mu_n = 2000$ and $\mu_p = 750$,

$$n_i = \frac{10^{-5}}{q(2750)} = 2.27 \times 10^{10} \text{ cm}^{-3}.$$

Solving the silicon $n_i(T)$ equation gives $T \approx 316.6 \text{ K}$. Below this temperature, intrinsic silicon is classified as an insulator by the table definition.

Problem 2.18 The conductor boundary is approximately $\sigma = 1000(\Omega \cdot \text{cm})^{-1}$. With $\mu_n = 100$ and $\mu_p = 50$,

$$n_i = \frac{1000}{q(150)} = 4.16 \times 10^{19} \text{ cm}^{-3}.$$

Solving $n_i(T)$ for silicon gives $T \approx 2701 \text{ K}$. Since silicon melts near 1430 K , solid silicon cannot realistically reach this condition.

Problem 2.19 A donor has one extra valence electron, while an acceptor has one missing bond electron. If one donor and one acceptor occupy adjacent lattice positions, the donor electron can fill the acceptor vacancy. Thus the pair is compensated. The result is no additional free electron or hole, except the thermally generated intrinsic carriers.

Problem 2.20 Germanium is group IV. Donor atoms are generally group V elements, and acceptor atoms are generally group III elements.

$$\text{Donors: } N, P, As, Sb, Bi; \quad \text{Acceptors: } B, Al, Ga, In, Tl.$$

Problem 2.21 GaAs is a III-V compound: Ga is group III and As is group V. Silicon is group IV.

(a) Si replacing Ga has one extra valence electron relative to Ga, so it behaves as a **donor**.

(b) Si replacing As has one fewer valence electron relative to As, so it behaves as an **acceptor**.

Problem 2.22 InP is a III-V compound: In is group III and P is group V. Germanium is group IV.

(a) Ge replacing In has one extra valence electron, so it behaves as a **donor**.

(b) Ge replacing P has one fewer valence electron, so it behaves as an **acceptor**.

Problem 2.23 Use $j = \sigma E = E/\rho$. Therefore

$$E = j\rho = (10000)(0.02) = 200 \text{ V/cm}.$$

Problem 2.24 For n-type material, $n \approx N_D = 10^{17} \text{ cm}^{-3}$. Thus

$$j_{max} = qnv_{max} = (1.602 \times 10^{-19})(10^{17})(10^7) = 1.60 \times 10^5 \text{ A/cm}^2.$$

Problem 2.25 The region volume is

$$V = (0.5\text{ }\mu\text{m})(5\text{ }\mu\text{m})(0.5\text{ }\mu\text{m}) = 1.25\text{ }\mu\text{m}^3 = 1.25 \times 10^{-12} \text{ cm}^3.$$

The number of boron atoms is

$$N = (10^{16})(1.25 \times 10^{-12}) = 1.25 \times 10^4.$$

1.25×10^4 **boron atoms**.

Problem 2.26 Arsenic is a donor, so $N_D = 3 \times 10^{17} \text{ cm}^{-3}$, $N_A = 0$. The silicon is **n-type**. At 300 K, $n_i \approx 10^{10} \text{ cm}^{-3}$. Since $N_D \gg 2n_i$,

$$n \approx N_D = 3 \times 10^{17} \text{ cm}^{-3}, \quad p = \frac{n_i^2}{n} = \frac{10^{20}}{3 \times 10^{17}} = 333 \text{ cm}^{-3}.$$

At 250 K, $n_i^2 \approx 4.53 \times 10^{15} \text{ cm}^{-6}$:

$$n \approx 3 \times 10^{17} \text{ cm}^{-3}, \quad p = \frac{4.53 \times 10^{15}}{3 \times 10^{17}} = 0.0151 \text{ cm}^{-3}.$$

Problem 2.27 Boron is an acceptor, so $N_A = 6 \times 10^{18} \text{ cm}^{-3}$, $N_D = 0$. The material is **p-type**. At 300 K,

$$p \approx 6 \times 10^{18} \text{ cm}^{-3}, \quad n = \frac{10^{20}}{6 \times 10^{18}} = 16.7 \text{ cm}^{-3}.$$

At 200 K, $n_i^2 \approx 5.28 \times 10^9 \text{ cm}^{-6}$:

$$p \approx 6 \times 10^{18} \text{ cm}^{-3}, \quad n = \frac{5.28 \times 10^9}{6 \times 10^{18}} = 8.80 \times 10^{-10} \text{ cm}^{-3}.$$

Problem 2.28 Arsenic is donor and boron is acceptor:

$$N_D = 2 \times 10^{18}, \quad N_A = 8 \times 10^{18} \text{ cm}^{-3}.$$

Since $N_A > N_D$, the sample is **p-type**. The net acceptor concentration is

$$p \approx N_A - N_D = 6 \times 10^{18} \text{ cm}^{-3}.$$

The minority concentration is

$$n = \frac{n_i^2}{p} = \frac{10^{20}}{6 \times 10^{18}} = 16.7 \text{ cm}^{-3}.$$

Problem 2.29 Boron is acceptor and phosphorus is donor:

$$N_A = 5 \times 10^{17}, \quad N_D = 2 \times 10^{17} \text{ cm}^{-3}.$$

The material is **p-type**. Then

$$p \approx N_A - N_D = 3 \times 10^{17} \text{ cm}^{-3}, \quad n = \frac{10^{20}}{3 \times 10^{17}} = 333 \text{ cm}^{-3}.$$

Problem 2.30 Given $N_D = 10^{16}$, $N_A = 5 \times 10^{16}$, and $n_i = 10^{11} \text{ cm}^{-3}$. Since $N_A > N_D$, the sample is p-type:

$$p \approx N_A - N_D = 4 \times 10^{16} \text{ cm}^{-3}.$$

Using $np = n_i^2$,

$$n = \frac{(10^{11})^2}{4 \times 10^{16}} = 2.50 \times 10^5 \text{ cm}^{-3}.$$

Problem 2.31 Given $N_A = 10^{15}$, $N_D = 10^{14}$, and $n_i = 5 \times 10^{13} \text{ cm}^{-3}$. Let

$$\Delta = N_A - N_D = 9 \times 10^{14} \text{ cm}^{-3}.$$

For p-type material,

$$p = \frac{\Delta + \sqrt{\Delta^2 + 4n_i^2}}{2} = \frac{9 \times 10^{14} + \sqrt{(9 \times 10^{14})^2 + 4(5 \times 10^{13})^2}}{2} = 9.03 \times 10^{14} \text{ cm}^{-3}.$$

Then

$$n = \frac{n_i^2}{p} = \frac{2.5 \times 10^{27}}{9.03 \times 10^{14}} = 2.77 \times 10^{12} \text{ cm}^{-3}.$$

Problem 2.32 Given $N_A = 2 \times 10^{17}$, $N_D = 3 \times 10^{17}$, and $n_i = 10^{17} \text{ cm}^{-3}$. Let

$$\Delta = N_D - N_A = 10^{17} \text{ cm}^{-3}.$$

For n-type material,

$$n = \frac{\Delta + \sqrt{\Delta^2 + 4n_i^2}}{2} = \frac{10^{17} + \sqrt{(10^{17})^2 + 4(10^{17})^2}}{2} = 1.62 \times 10^{17} \text{ cm}^{-3}.$$

Then

$$p = \frac{n_i^2}{n} = \frac{10^{34}}{1.62 \times 10^{17}} = 6.18 \times 10^{16} \text{ cm}^{-3}.$$

Problem 2.33 Donor concentration: $N_D = 5 \times 10^{16} \text{ cm}^{-3}$, $N_A = 0$, so the material is **n-type**. At 300 K,

$$n \approx 5 \times 10^{16} \text{ cm}^{-3}, \quad p = \frac{10^{20}}{5 \times 10^{16}} = 2.0 \times 10^3 \text{ cm}^{-3}.$$

From the mobility approximation at $N_T = 5 \times 10^{16} \text{ cm}^{-3}$,

$$\mu_n \approx 987 \text{ cm}^2/(\text{V} \cdot \text{s}), \quad \mu_p \approx 291 \text{ cm}^2/(\text{V} \cdot \text{s}).$$

Thus

$$\sigma = q(\mu_n n + \mu_p p) \approx (1.602 \times 10^{-19})(987)(5 \times 10^{16}) = 7.91(\Omega \cdot \text{cm})^{-1},$$

$$\rho = \frac{1}{\sigma} = 0.126 \Omega \cdot \text{cm}.$$

Problem 2.34 Acceptor concentration: $N_A = 2.5 \times 10^{18} \text{ cm}^{-3}$, $N_D = 0$, so the material is **p-type**.

$$p \approx 2.5 \times 10^{18} \text{ cm}^{-3}, \quad n = \frac{10^{20}}{2.5 \times 10^{18}} = 40 \text{ cm}^{-3}.$$

At $N_T = 2.5 \times 10^{18} \text{ cm}^{-3}$,

$$\mu_n \approx 173 \text{ cm}^2/(\text{V} \cdot \text{s}), \quad \mu_p \approx 73.7 \text{ cm}^2/(\text{V} \cdot \text{s}).$$

Therefore

$$\rho \approx \frac{1}{q\mu_p p} = \frac{1}{(1.602 \times 10^{-19})(73.7)(2.5 \times 10^{18})} = 3.39 \times 10^{-2} \Omega \cdot \text{cm}.$$

Problem 2.35 Indium is group III, so in silicon it is an **acceptor**. Thus $N_A = 8 \times 10^{19} \text{ cm}^{-3}$ and the material is **p-type**.

$$p \approx 8 \times 10^{19} \text{ cm}^{-3}, \quad n = \frac{10^{20}}{8 \times 10^{19}} = 1.25 \text{ cm}^{-3}.$$

At $N_T = 8 \times 10^{19}$,

$$\mu_n \approx 95.7, \quad \mu_p \approx 49.9 \text{ cm}^2/(\text{V} \cdot \text{s}).$$

Then

$$\rho \approx \frac{1}{q\mu_p p} = \frac{1}{(1.602 \times 10^{-19})(49.9)(8 \times 10^{19})} = 1.56 \times 10^{-3} \Omega \cdot \text{cm}.$$

Problem 2.36 Phosphorus is donor and boron is acceptor:

$$N_D = 4.5 \times 10^{16}, \quad N_A = 5.5 \times 10^{16} \text{ cm}^{-3}.$$

Since $N_A > N_D$, the material is **p-type**. The majority and minority concentrations are

$$p \approx N_A - N_D = 10^{16} \text{ cm}^{-3}, \quad n = \frac{10^{20}}{10^{16}} = 10^4 \text{ cm}^{-3}.$$

The total impurity concentration is $N_T = N_A + N_D = 10^{17} \text{ cm}^{-3}$. From Fig. 2.8,

$$\mu_n \approx 800, \quad \mu_p \approx 230 \text{ cm}^2/(\text{V} \cdot \text{s}).$$

Thus

$$\rho \approx \frac{1}{q\mu_p p} = \frac{1}{(1.602 \times 10^{-19})(230)(10^{16})} = 2.7 \Omega \cdot \text{cm}.$$

Problem 2.37 Example 2.5 asks for the doping required to obtain $\rho = 0.054\Omega \cdot \text{cm}$, using graph-based mobility. For p-type material with only acceptors,

$$\sigma = \frac{1}{\rho} = 18.52(\Omega \cdot \text{cm})^{-1}, \quad \sigma \approx q\mu_p N_A.$$

Thus the required product is

$$\mu_p N_A = \frac{1}{q\rho} = 1.16 \times 10^{20} (\text{V} \cdot \text{s} \cdot \text{cm})^{-1}.$$

Using the mobility graph iteratively, a suitable value is approximately

$$N_A \approx 1.3 \times 10^{18} \text{ cm}^{-3}, \quad \mu_p \approx 89 \text{ cm}^2/(\text{V} \cdot \text{s}),$$

which gives $\mu_p N_A \approx 1.16 \times 10^{20}$. $N_A \approx 1.3 \times 10^{18} \text{ cm}^{-3}$.

Problem 2.38 Repeat Example 2.5 using the mobility approximation equation from Fig. 2.8:

$$\mu_n = 92 + \frac{1270}{1 + (N_T/1.3 \times 10^{17})^{0.91}}.$$

For n-type material with only donors, $N_T = N_D$ and

$$\mu_n(N_D)N_D = \frac{1}{q\rho} = 1.16 \times 10^{20}.$$

Solving iteratively gives

$$\boxed{N_D \approx 1.84 \times 10^{17} \text{ cm}^{-3}}, \quad \mu_n \approx 627 \text{ cm}^2/(\text{V} \cdot \text{s}).$$

This is close to the graphical result $N_D \approx 2 \times 10^{17} \text{ cm}^{-3}$.

Problem 2.39 Repeat Problem 2.37 using the hole mobility approximation

$$\mu_p = 48 + \frac{447}{1 + (N_T/6.3 \times 10^{16})^{0.76}}.$$

For p-type silicon with only acceptors, $N_T = N_A$ and

$$\mu_p(N_A)N_A = \frac{1}{q\rho} = 1.16 \times 10^{20}.$$

Solving gives

$$\boxed{N_A \approx 1.30 \times 10^{18} \text{ cm}^{-3}}, \quad \mu_p \approx 88.6 \text{ cm}^2/(\text{V} \cdot \text{s}).$$

Problem 2.40 For p-type silicon with only acceptors and $\rho = 0.5\Omega \cdot \text{cm}$,

$$\mu_p(N_A)N_A = \frac{1}{q\rho} = \frac{1}{(1.602 \times 10^{-19})(0.5)} = 1.25 \times 10^{19}.$$

Using the mobility equation for holes gives

$$\boxed{N_A \approx 4.0 \times 10^{16} \text{ cm}^{-3}}.$$

Problem 2.41 Yes. If nearly equal amounts of donors and acceptors are added, the material becomes compensated: $N_A \approx N_D$, so $n \approx p \approx n_i$. However, the ionized impurities still scatter carriers, reducing μ_n and μ_p . Since

$$\rho = \frac{1}{q(\mu_n n + \mu_p p)},$$

reduced mobility can make extrinsic compensated silicon more resistive than intrinsic silicon. It is not intrinsic because it contains impurities.

Problem 2.42 For n-type silicon with only donor impurities and $\rho = 3.0\Omega \cdot \text{cm}$,

$$\mu_n(N_D)N_D = \frac{1}{q\rho} = 2.08 \times 10^{18}.$$

Solving with the electron mobility equation gives

$$N_D \approx 1.55 \times 10^{15} \text{ cm}^{-3}.$$

Problem 2.43 A conductor requires roughly $\rho \leq 10^{-3} \Omega \cdot \text{cm}$.

(a) For n-type silicon,

$$\mu_n(N_D)N_D = \frac{1}{q(10^{-3})} = 6.24 \times 10^{21}.$$

Solving gives $N_D \approx 6.5 \times 10^{19} \text{ cm}^{-3}$.

(b) For p-type silicon,

$$\mu_p(N_A)N_A = 6.24 \times 10^{21}.$$

Solving gives $N_A \approx 1.26 \times 10^{20} \text{ cm}^{-3}$.

Problem 2.44 Here $N_D = N_A = 5.0 \times 10^{19} \text{ cm}^{-3}$. The net doping is zero, so charge neutrality gives approximately $n = p = n_i$. But the total impurity concentration is very high:

$$N_T = N_A + N_D = 10^{20} \text{ cm}^{-3}.$$

Using mobility equations, $\mu_n \approx 95$ and $\mu_p \approx 50 \text{ cm}^2/(\text{V} \cdot \text{s})$. Therefore

$$\rho = \frac{1}{qn_i(\mu_n + \mu_p)} \approx \frac{1}{(1.602 \times 10^{-19})(10^{10})(145)} = 4.3 \times 10^6 \Omega \cdot \text{cm}.$$

This is an **insulator** by resistivity. It is **not intrinsic**; it is heavily compensated extrinsic silicon.

Problem 2.45 Initial p-type wafer: $\rho = 1 \Omega \cdot \text{cm}$, only boron. Using the hole mobility equation,

$$\mu_p(N_A)N_A = \frac{1}{q(1)} = 6.24 \times 10^{18},$$

so initially

$$N_{A0} \approx 1.66 \times 10^{16} \text{ cm}^{-3}.$$

For target $\rho = 0.25 \Omega \cdot \text{cm}$ with extra acceptors,

$$\mu_p(N_A)N_A = \frac{1}{q(0.25)} = 2.50 \times 10^{19},$$

giving

$$N_{A,f} \approx 1.12 \times 10^{17} \text{ cm}^{-3}.$$

Additional acceptor concentration:

$$\Delta N_A = N_{A,f} - N_{A0} = 9.5 \times 10^{16} \text{ cm}^{-3}.$$

If donors are added instead, the sample must cross compensation and become n-type. Solving

$$\mu_n(N_D + N_{A0})(N_D - N_{A0}) = \frac{1}{q(0.25)}$$

gives approximately $N_D \approx 4.3 \times 10^{16} \text{ cm}^{-3}$; the result is **n-type**.

Problem 2.46 The wafer is doped with phosphorus, so $N_D = 10^{16} \text{ cm}^{-3}$ and the material is n-type.

(a) From Fig. 2.8, $\mu_n \approx 1250 \text{ cm}^2/(\text{V} \cdot \text{s})$. Thus

$$\sigma \approx q\mu_n N_D = (1.602 \times 10^{-19})(1250)(10^{16}) = 2.00(\Omega \cdot \text{cm})^{-1}.$$

(b) Add boron until the conductivity is $4.0(\Omega \cdot \text{cm})^{-1}$. The material must become p-type after compensation. Solve

$$\mu_p(N_A + 10^{16})(N_A - 10^{16}) = \frac{4.0}{q} = 2.50 \times 10^{19}.$$

Using the mobility equation gives $N_A \approx 1.32 \times 10^{17} \text{ cm}^{-3}$.

Problem 2.47 The wafer has background boron concentration $N_A = 10^{16} \text{ cm}^{-3}$.

(a) From Fig. 2.8, $\mu_p \approx 406 \text{ cm}^2/(\text{V} \cdot \text{s})$, so

$$\sigma \approx q\mu_p N_A = (1.602 \times 10^{-19})(406)(10^{16}) = 0.651(\Omega \cdot \text{cm})^{-1}.$$

(b) Add phosphorus until $\sigma = 5.5(\Omega \cdot \text{cm})^{-1}$. The final material becomes n-type. Solve

$$\mu_n(N_D + 10^{16})(N_D - 10^{16}) = \frac{5.5}{q} = 3.43 \times 10^{19}.$$

This gives approximately $N_D \approx 4.6 \times 10^{16} \text{ cm}^{-3}$.

Problem 2.48 The thermal voltage is

$$V_T = \frac{kT}{q} = 8.62 \times 10^{-5} T \text{ V}.$$

$T(\text{K})$	50	75	100	150	200	250	300	350	400
$V_T(\text{mV})$	4.31	6.46	8.62	12.9	17.2	21.6	25.9	30.2	34.5

Problem 2.49 From Fig. P2.49, n decreases linearly from 10^{18} cm^{-3} to 0 across

$$W_B = 0.5 \mu\text{m} = 5 \times 10^{-5} \text{ cm}.$$

Thus

$$\frac{dn}{dx} = \frac{0 - 10^{18}}{5 \times 10^{-5}} = -2.0 \times 10^{22} \text{ cm}^{-4}.$$

The electron diffusion coefficient is

$$D_n = \mu_n V_T = 350(0.025) = 8.75 \text{ cm}^2/\text{s}.$$

Electron diffusion current density:

$$j_{n,diff} = qD_n \frac{dn}{dx} = (1.602 \times 10^{-19})(8.75)(-2.0 \times 10^{22}) = -2.80 \times 10^4 \text{ A/cm}^2.$$

$$j_{n,diff} = -28.0 \text{ kA/cm}^2.$$

Problem 2.50 Given

$$p(x) = 10^5 + 10^{19} \exp\left(-\frac{x}{L_p}\right), \quad L_p = 2.0 \mu\text{m} = 2 \times 10^{-4} \text{ cm}.$$

Then

$$\frac{dp}{dx} = -\frac{10^{19}}{L_p} \exp\left(-\frac{x}{L_p}\right).$$

For holes,

$$j_{p,diff} = -qD_p \frac{dp}{dx} = qD_p \frac{10^{19}}{L_p} \exp\left(-\frac{x}{L_p}\right).$$

With $D_p = 15 \text{ cm}^2/\text{s}$,

$$j_{p,diff}(x) = (1.602 \times 10^{-19})(15) \frac{10^{19}}{2 \times 10^{-4}} \exp\left(-\frac{x}{2 \mu\text{m}}\right) = 1.20 \times 10^5 e^{-x/(2 \mu\text{m})} \text{ A/cm}^2.$$

At $x = 0$, $j = 1.20 \times 10^5 \text{ A/cm}^2$. For $A = 10 \mu\text{m}^2 = 10^{-7} \text{ cm}^2$,

$$I(0) = jA = (1.20 \times 10^5)(10^{-7}) = 1.20 \times 10^{-2} \text{ A} = 12.0 \text{ mA}.$$

Problem 2.51 At thermal equilibrium, the total hole current must be zero:

$$j_p = q\mu_p pE - qD_p \frac{dp}{dx} = 0.$$

Using $D_p/\mu_p = V_T$,

$$E = V_T \frac{1}{p} \frac{dp}{dx}.$$

The acceptor profile is approximately

$$p(x) \approx N_A(x) = 10^{14} + 10^{18}e^{-10^4 x}, \quad x \text{ in cm.}$$

Therefore

$$\frac{dp}{dx} = -10^{22}e^{-10^4 x},$$

and

$$E(x) = 0.025 \frac{-10^{22}e^{-10^4 x}}{10^{14} + 10^{18}e^{-10^4 x}}.$$

At $x = 0$,

$$E(0) = 0.025 \frac{-10^{22}}{10^{14} + 10^{18}} \approx -250 \text{ V/cm.}$$

At $x = 5 \mu\text{m} = 5 \times 10^{-4} \text{ cm}$,

$$E(5 \mu\text{m}) \approx 0.025 \frac{-10^{22}e^{-5}}{10^{14} + 10^{18}e^{-5}} \approx -246 \text{ V/cm.}$$

The internal electric field is nonzero because it must cancel diffusion current.

Problem 2.52 The electric field arrow points left, so take $E = -20 \text{ V/cm}$. Use $\mu_n = 350$, $\mu_p = 150 \text{ cm}^2/(\text{V} \cdot \text{s})$, $D_n = \mu_n V_T = 8.75 \text{ cm}^2/\text{s}$, and $D_p = \mu_p V_T = 3.75 \text{ cm}^2/\text{s}$. At $x = 0$,

$$p = 1.01 \times 10^{18}, \quad n = 10^{16} \text{ cm}^{-3}.$$

The concentration gradients are approximately

$$\frac{dn}{dx} = \frac{10^4 - 10^{16}}{2 \times 10^{-4}} = -5.0 \times 10^{19} \text{ cm}^{-4},$$

$$\frac{dp}{dx} = \frac{10^{18} - 1.01 \times 10^{18}}{2 \times 10^{-4}} = -5.0 \times 10^{19} \text{ cm}^{-4}.$$

Components at $x = 0$:

$$j_{n,drift} = q\mu_n n E = (1.602 \times 10^{-19})(350)(10^{16})(-20) = -11.2 \text{ A/cm}^2,$$

$$j_{p,drift} = q\mu_p p E = (1.602 \times 10^{-19})(150)(1.01 \times 10^{18})(-20) = -485 \text{ A/cm}^2,$$

$$j_{n,diff} = qD_n \frac{dn}{dx} = (1.602 \times 10^{-19})(8.75)(-5.0 \times 10^{19}) = -70.1 \text{ A/cm}^2,$$

$$j_{p,diff} = -qD_p \frac{dp}{dx} = -(1.602 \times 10^{-19})(3.75)(-5.0 \times 10^{19}) = +30.0 \text{ A/cm}^2.$$

Thus

$$j_T \approx -11.2 - 485 - 70.1 + 30.0 = -536 \text{ A/cm}^2.$$

At $x = 1.0 \mu\text{m}$, using midpoint carrier densities gives a total close to $-5.3 \times 10^2 \text{ A/cm}^2$; the diffusion components remain the same if the profiles are treated as linear.

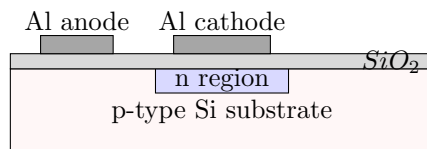
Problem 2.53 For $N_A > N_D$ with two acceptors for each donor, donor electrons fill only part of the acceptor states. The remaining uncompensated acceptors create holes, so the material is p-type. In an energy-band diagram, acceptor levels E_A lie just above the valence band E_V , donor levels E_D lie below the conduction band E_C , and the Fermi level is closer to E_V than to E_C .

Problem 2.54 A photon can create an electron-hole pair if $E = hc/\lambda \geq E_G$. For silicon $E_G \approx 1.12 \text{ eV}$:

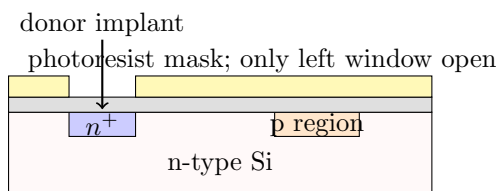
$$\lambda_{max} = \frac{hc}{E_G} = \frac{(6.626 \times 10^{-34} \text{ J} \cdot \text{s})(3.0 \times 10^8 \text{ m/s})}{(1.12)(1.602 \times 10^{-19} \text{ J})} = 1.108 \times 10^{-6} \text{ m.}$$

$$\lambda_{max} \approx 1.108 \mu\text{m}.$$

Problem 2.55 If the substrate is p-type, the diffused region must be n-type to form a pn diode. The p-side is the anode and the n-side is the cathode.



Problem 2.56 The extra n^+ step should be inserted after opening the contact window over the n-type region and before aluminum deposition. Photoresist can mask all regions except the left-hand n-contact window, so donor implantation occurs only there.



Problem 2.57 For diamond-cubic silicon:

(a) Atom count:

$$8 \left(\frac{1}{8} \right) + 6 \left(\frac{1}{2} \right) + 4 = 1 + 3 + 4 = 8 \text{ atoms.}$$

(b) Lattice constant $a = 0.543 \text{ nm} = 0.543 \times 10^{-7} \text{ cm}$, so

$$V = a^3 = (0.543 \times 10^{-7})^3 = 1.60 \times 10^{-22} \text{ cm}^3.$$

(c) Atomic density:

$$N = \frac{8}{1.60 \times 10^{-22}} = 5.00 \times 10^{22} \text{ atoms/cm}^3.$$

(d) Mass of one cell:

$$m = (2.33 \text{ g/cm}^3)(1.60 \times 10^{-22} \text{ cm}^3) = 3.73 \times 10^{-22} \text{ g.}$$

(e) Silicon atomic mass is about 28.086 nucleons per atom. One unit cell has 8 atoms, so

$$m_{p/n} \approx \frac{3.73 \times 10^{-22}}{8(28.086)} = 1.66 \times 10^{-24} \text{ g.}$$

This is reasonable because it is close to the known proton/neutron mass.